

CSCI [404](#) (Sec. 1) [001184]
Design and Analysis of Algorithms

CSCI [504](#) (Sec. 1) [001185]
Design and Analysis of Algorithms

Fall 2026 SYLLABUS

INSTRUCTOR: Andrew A. Anda, Ph.D., Professor of Computer Science
OFFICE: CH-366AE
Telephone: (320) 308-2044
E-MAIL: aanda@stcloudstate.edu
(please use "CSCI404:" or "CSCI504" in the Subject field)

Instructor Schedule:
<https://aanda-stcloudstate-edu.github.io/WebSpace/Wkly-sched-f26.pdf>

Student Office Hours Zoom: <https://minnstate.zoom.us/j/7680194778>

{I sometimes won't be available for a Student Support office hour - this will usually be because I'm attending a meeting}

(and by appointment)

CLASS TIME AND LOCATION:
CSCI 404/504 Class: MWF: 14:00 - 14:50 in CH 9

Final Exam: Dec. 7, M: 12:20 - 14:35 in TBD

REQUIRED TEXT: * Introduction to Algorithms (4th Ed.), by Thomas Cormen, et al, The MIT Press 2022

COURSE DESCRIPTION:
Computing time functions; maximum, minimum and average computing time of various algorithms.

STUDENT LEARNING OUTCOMES:

- * Construct software systems of varying complexity.
- * Model computer based systems considering trade-offs involved in design choices.
- * Analyze a problem and identify the computing requirements appropriate for its solution.
- * Apply knowledge of computing and mathematics to algorithm design.
- * Design, implement, and evaluate an algorithm to meet desired needs.
- * Apply mathematical foundations, algorithmic principles, and computer science theory to the modeling and design of computer-based systems in a way that demonstrates comprehension of the trade-offs involved in design choices.
- * Use current techniques, skills, and tools necessary for computing practice.

GRADUATE 504 SLOs:

* Compare theoretical expectations with practical implementation results from cutting edge algorithms and evaluate their effectiveness.

COURSE OUTLINE (not necessarily in order of coverage):

Algorithms in computing. 10%
Analyzing and designing algorithms. 10%
Run times. 10%
Sort algorithms. 15%
Data structures. 10%
Dynamic programming. 15%
Greedy algorithms. 10%
Graph algorithms. 10%
Divide and conquer. 10%

PREREQUISITES:

CSCI331 - Software Systems
AND MATH271 - Discrete Mathematics

ACADEMIC HONESTY:

You are expected to do your own homework. If you copy someone else's work or allow someone else to copy your work, you are being academically dishonest and will be subject to severe disciplinary action which may include any or all of: no credit for the work in question, a failing grade

for the course, notification to the university that you have violated your

Code of Conduct. Use of recording devices during exams is prohibited. If you must quote or paraphrase another source, citation is essential, otherwise plagiarism has been committed. You are expected to be familiar with your rights and obligations as outlined in the "Code of Conduct"

[<http://www.stcloudstate.edu/studenthandbook/code>] See also:
[<http://www.stcloudstate.edu/studenthandbook/policies/cheating.asp>]

[http://www.stcloudstate.edu/studenthandbook/academic_policies/honesty.asp]

DECORUM: Conduct yourself so as not to distract others. In particular

- * Turn off all cell phones, watch alarms, and similar devices.
- * Do not engage in distracting activities like reading a newspaper or sleeping. (some of you snore...)
- * Arrive for class on time, and be ready to begin.
If you must be late, be quiet.
- * If you are early, sit toward the middle of a row.
- * Do not start packing up before class ends.

ATTENDANCE:

You are responsible for knowing what happens at each class meeting.

EXAMS:

There will be a small set of intermediate exams and a comprehensive final exam.

Exams will be based on the notes; that is, on the presentations in class as well as on the assigned readings and assignments.

Calculators will be neither necessary nor allowed except by prior arrangement.

Make-up exams will be given in only in extreme (or university sanctioned) circumstances and only with prior notice.

GRADING:

Your grade will be determined by the contributions of your scores.

APPROXIMATE POINTS: TBD

Assignments	TBD%
Intermediate Exams	TBD%
Comprehensive Final Exam	TBD%

CAVEAT:

I reserve the right to amend the contents of this syllabus with notification.

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Student accessibility Services

Student Accessibility Services maintains Accessibility Syllabus Statements for faculty to include in their course syllabi. Check out the Student Accessibility Services [Faculty and Staff Resources webpage](#) for additional information statements, the accommodation process, and more. The below syllabus statement sample is supported by Student Accessibility Services:

Accessibility Statement:

This course is designed with accessibility in mind and aims to be inclusive to all students. If you encounter barriers to accessing course materials, participating fully, or demonstrating your learning, please don't hesitate to reach out. I am committed to working with you to ensure equitable access. Students who have a disability, or who think they may benefit from formal accommodations, are encouraged to contact Student Accessibility Services (SAS) as early as possible. SAS is located at Centennial Hall 210 and can be reached at 320-308-4080 or sas@stcloudstate.edu. For more information or to request accommodations, please visit www.stcloudstate.edu/sas.

EAB Navigate Student Success Tool

- St. Cloud State University uses the Navigate platform, which allows faculty and advisors to provide proactive and coordinated care for students. Throughout the semester, I may submit updates on students with attendance or academic concerns so that advisors can reach out to provide additional support and/or connect you to campus resources. I may also contact you through email, before or after class, or encourage you to meet with me during my office hours. Please know I'm here to support you in your academic success. You can learn more about the Navigate tool at [Navigate | St. Cloud State University](#)

Student Support Hours (Office Hours):

- **Student Support Hours are dedicated times when I am available to support you and your success.** During these hours, I am fully available to meet with you, whether to discuss course content, assignments, academic planning, or other questions you may have. I may be working on other tasks, but if you arrive, that time becomes student focused.
- Student Support Hours may be held in person or virtually, depending on the course format. If you are unable to meet during the scheduled times, I am happy to arrange an alternative meeting when possible. You are also welcome to stop by at other times if my door is open; if I am unavailable at that moment, we can schedule a time to connect.
- Think of these hours as an open invitation to engage - no appointment, no pressure, and no staring sadly out the window required.

Syllabus statement for faculty who are supportive of exploratory use of AI

- Artificial Intelligence (AI) is a tool available to the public. Within this course, you are expected to submit assignments and discussions that are your original thought. Within this course, you are encouraged to explore how AI may impact routine tasks within our profession. The use of AI for brainstorming, outlining, or creating support for your academic assignments is allowable in this course. Submitting AI-generated assignments as your own original thought is considered to breach our Academic Integrity practices and should not occur. Please ask me if you are unsure of where AI and Academic Integrity intersect in this course. Please refer to the Academic Integrity policies that apply to all courses within our institution.
- Supported tools such as Copilot and Adobe AI tools may be used in this course for limited academic tasks such as idea generation, outlining, or early-stage revisions. However, all submitted work must reflect your own understanding, voice, and academic reasoning.

AI tools should serve as a support, not a replacement, for your thinking. Submitting work that is primarily AI-generated without acknowledgment is a breach of our Academic Integrity Policy.

If you use AI tools as part of your process, you must be transparent about how they were used (e.g., "Used Copilot to draft bullet points; expanded and revised with personal analysis"). When in doubt, ask for clarification before submitting your work.

Please consult the Academic Integrity Policy for more information and let's maintain an open dialogue about ethical and responsible technology use.

Our Land Commitment

St. Cloud State University is committed to developing a land acknowledgment that is not just words but actions. We will actively promote education among our faculty, staff, and students on Indigenous issues, including research on the historical and ongoing harms impacting our Indigenous communities. We will also clearly outline our goals to make a positive and lasting contribution to efforts to build relationships with the Tribal Nations represented within our local and broader community. St. Cloud State University commits to creating a substantive and empathic land acknowledgment, demonstrating our continued commitment to action. Faculty are encouraged to learn more about **Our Land Commitment** on the [OEA Land Commitment webpage](#) for the most up to date information.

SAFETY:

What to do if you have experienced sexual assault, rape, domestic violence, sexual harassment or stalking:

Please read this resource:

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[<https://www5.stcloudstate.edu/Policies/SCSU/Viewer.aspx?id=60>]

Your professors fulfil the role of mandatory reporters. If you want to speak confidentially, see the above link for a list of confidential options.

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Peer Wellness Coaching

<https://www.stcloudstate.edu/healthwellness/get-healthy/peer-coaching.aspx>

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Accommodations Statement

St. Cloud State is an affirmative action, equal opportunity employer and educator.

SCSU is committed to a policy of nondiscrimination in employment and education opportunity and works to provide reasonable accommodations for all persons with disabilities.

Accommodations are provided on an individualized, as-needed basis, determined through appropriate documentation of need.

Please contact Student Disability Services, sds@stcloudstate.edu, or 320.308.4080, office CH 202 to meet and discuss reasonable and appropriate accommodations.

The accommodations authorized may be discussed with your instructor. All discussions will remain confidential.

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Violence Statement

In the event that you choose to write or speak about having survived sexualized violence, including rape, sexual assault, dating violence, domestic violence, or stalking, and specify that this violence occurred while you were an SCSU student, federal and state education laws require that, as your instructor, I notify the Title IX officer, Ellyn Bartges.

She (or her designee), will contact you to let you know about accommodations and support services at SCSU and possibilities for holding accountable the person who harmed you.

If you do not want the Title IX Officer notified, instead of disclosing this information to your instructor, you can speak confidentially with the following people on campus and in the community.

They can connect you with support services and discuss options for holding the perpetrator accountable.

SCSU's Gender Violence Prevention Program	320.308.3995	Lee LaDue
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Central Mn Sexual Assault Center (Community program)	320.251.4357	24-hour hotline
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SCSU's Counseling and Psychological Services	320-308-3171
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If you are a survivor or someone concerned about a survivor and need immediate information on what to do, please go to:
<http://www.stcloudstate.edu/womenscenter/>

Resource Links:

- [D2L Brightspace](#)
- [SCSU Student Resources for Online Learning](#)
- [SCSU Zoom Resources](#)
- [SCSU Student Information for Attending Classes Off-Campus](#)
- [SCSU: The Write Place](#)
- [SCSU Information Technology Services](#)
- [SCSU Medical Clinic](#)
- [SCSU Counseling and Psychological Services](#)
- [SCSU Library](#)
- [SCSU Student Code of Conduct](#)

Additional SCSU student resources, curated and compiled by the Academic Affairs office, are presented in: [Fall 2023 Student Instructional Resource and Support Guide](#)

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